

# RevenueGuard

Summary, technical statement and workflow.

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A short read: the problem in a paragraph, the technical statement in one sentence, the seven-stage control loop, how the recovery figure is measured, and the results.

**Rs 1.24 cr**

recovered per 2 days (mean of 8 seeds)

**38.4%**

of money the incidents put at risk

**0/8**

seeds where the router lost money

## The problem

Payment failures are almost never global. They are narrow — one gateway, one method, one issuing bank — and a narrow failure is invisible on the chart everyone watches. A slice carrying 4% of volume can collapse from 96% to 40% and move the headline success rate by about two points, well inside normal daily variation. Nobody pages, and the money leaves quietly until a human spots a pattern in a support queue.

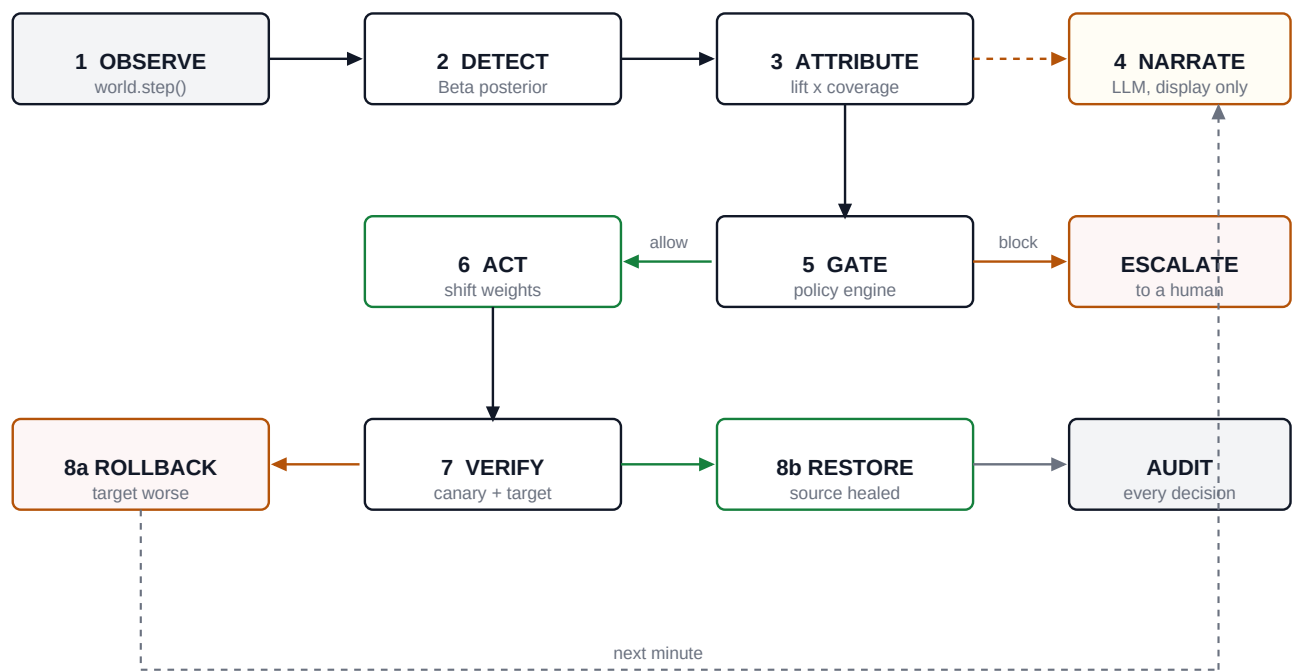
## The technical statement

**Detect per-slice success-rate degradation under two hostile statistical conditions — very low volume and very high test multiplicity — attribute the alarming population to a common dimension deterministically, and execute a bounded, reversible, fully audited traffic shift whose recovered revenue is measured against a control arm rather than projected.**

Each clause of that sentence is a design constraint, and each was chosen against a more obvious alternative:

| Constraint   | Obvious approach                                                       | What is done, and why                                                                                                                                              |
|--------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ground truth | Label synthetic rows, train a classifier, report precision and recall. | Outages are <i>scheduled</i> , so time-to-detect is latency against an event we caused. Training on self-generated labels measures the generator, not the world.   |
| Detection    | Threshold or z-score per slice.                                        | Beta posterior with cohort shrinkage. A slice with four attempts a minute swings 50–100% on noise; 207,360 tests over two days drown any fixed significance level. |
| Attribution  | Ask an LLM what is wrong.                                              | Lift with coverage, deterministic. A sampled token in the path of a money-moving action cannot be reproduced or defended.                                          |
| Action       | Reroute on detection.                                                  | A separate policy engine decides <i>permission</i> . Ten bounds, stopping rules, and every refusal recorded with its rule.                                         |
| Reversal     | Drain the bad gateway to zero.                                         | A 3% canary floor. A gateway with no traffic emits no observations, so you could never tell it recovered or route back.                                            |
| Measurement  | Count recoveries, multiply by ticket size.                             | Paired control and treatment on bit-identical demand, repeated across seeds, reported as a confidence interval.                                                    |

The workflow

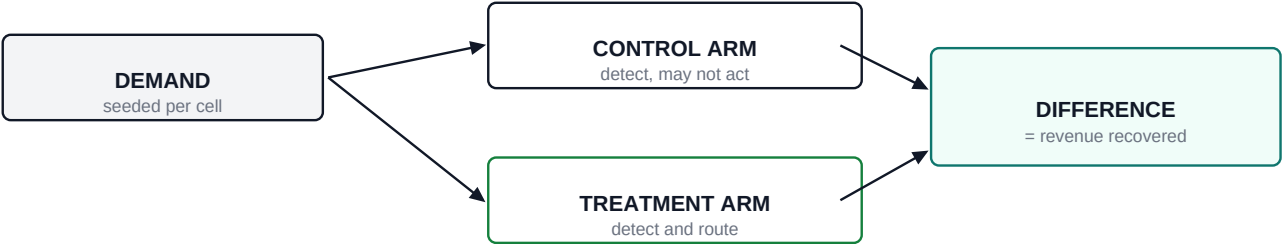


Seven stages, once a minute. The system sees only (slice, minute, attempts, successes) — never the incident plan that generated the traffic.

| Stage       | Guarantee it provides                                                                 |
|-------------|---------------------------------------------------------------------------------------|
| 1 Observe   | No ground truth is visible to any decision-making code.                               |
| 2 Detect    | Alarms only when the drop is both statistically real and materially large.            |
| 3 Attribute | Deterministic and reproducible; refuses to name a cause the evidence cannot support.  |
| 4 Narrate   | Display only. Cannot reach the ledger or any decision.                                |
| 5 Gate      | Ten explicit bounds; every verdict recorded with its rule, including refusals.        |
| 6 Act       | Bounded fraction, per key, with a canary always left behind.                          |
| 7 Verify    | Rolls back if the destination proved worse; restores in thirds once the source heals. |

# How the measurement works

One seed, one incident plan, one demand stream



**Guard: total attempts must match exactly across arms.**  
If they differ the run refuses to report a recovery figure at all - the comparison would be invalid.  
Repeated across seeds; the headline is the confidence interval, not any single run.

# Where AI is used

**DETERMINISTIC**  
statistics, rules, arithmetic

detection - attribution - policy - routing - rollback - audit  
*Reproducible. Same input, same decision, forever.*

**LANGUAGE MODEL**  
prose only

restates a finished attribution for the operator console  
*Never decides. Never reaches the ledger. Falls back to a template.*

**Why the line sits here**  
A sampled token in the path of a money-moving action cannot be reviewed, reproduced, or defended.  
Two tests enforce the boundary: the prompt carries no raw traffic, and no decision-path module imports the narrator.

# Results

**Rs 1.24 cr**  
recovered per 2 days, mean of 8 paired seeds

**38.4%**  
of the money the incidents put at risk

**0/8**  
seeds where the router lost money

| Metric            | Mean       | SD        | Min        | Max        |
|-------------------|------------|-----------|------------|------------|
| Revenue recovered | Rs 1.24 cr | Rs 5.98 L | Rs 1.13 cr | Rs 1.31 cr |
| Share of exposure | 38.4%      | 1.9%      | 35.6%      | 41.0%      |
| Success rate gain | 0.578 pp   | 0.020     | 0.542      | 0.604      |
| Routing actions   | 103.1      | 5.7       | 98         | 114        |
| Rollbacks         | 39.2       | 4.1       | 34         | 47         |

95% CI on mean recovery: **Rs 1.19 cr to Rs 1.29 cr**, which excludes zero. Cross-checked independently: measured recovery Rs 1.23 cr against an exposure reduction of Rs 1.32 cr — different quantities, agreeing within 1.5%.

# What is deliberately not claimed

- **The world is simulated.** Gateway health, demand and incidents are ours; the detector's view of them is the part under evaluation.
- **Gateway health does not degrade under load.** A real acquirer would push back as traffic arrives, so the recovery figure is an upper bound. This is the first item on the roadmap.
- **Routing weights cannot be set through Razorpay's public API.** Acquirer selection is Razorpay's own product. The executor demonstrates the half that is real — test-mode Orders carrying the decision context in their notes.
- **38% of exposure, not 90%.** Detection latency and deliberate policy caps set the ceiling.
- **About 39 rollbacks per 103 actions** are reported rather than tuned away.

Next

| # | Item                                           | Why it matters                                                                                                                                                                                                                     |
|---|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Cost of action                                 | Rerouting is not free: acquirer contracts carry volume commitments and per-transaction pricing differences. Netting that against recovered revenue turns a gross figure into a net one.                                            |
| 2 | Varied incident plans across seeds             | The current interval covers traffic randomness only. Randomising which incidents occur, and their depth and timing, would widen it honestly and make it a stronger claim.                                                          |
| 3 | A measured congestion curve                    | The capacity model's knee and slope are a plausible shape, not an observed one, and that curve is what sets the 80% shift cap. Fitting it to real acquirer telemetry would move the cap and is the most load-bearing unknown left. |
| 4 | Sensitivity analysis over the remaining bounds | The shift cap has now been measured and moved. The other nine are still defended by argument. The same sweep applied to each would show which earn their place.                                                                    |

Full detail in the companion document, *RevenueGuard — Complete Project Documentation*.